HOW TO PLAY

Powerhouse

The ATP clicker that runs on your Anki reviews.



1

Review like normal

Every answer that counts makes ATP for Mito, your mitochondrion.

2

Build pathways

Spend ATP on metabolism, from glycolysis to ATP synthase, so every future card is worth more.

3

Open vesicles

Loot drops while you review: ATP, 48 enzymes to collect and outfits for Mito.

4

Come back daily

Daily goal, streak, quests and 56 trophies. Later, Mito grows and divides.

Powerhouse only reads your review history. It never changes your cards, scheduling, decks or sync.

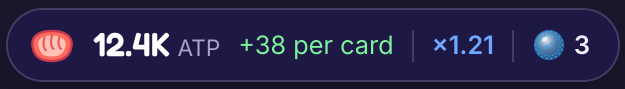
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GETTING AROUND

Three places in Anki

You never have to open the game to earn ATP. It counts your reviews in the background and shows up in three spots.

The review pill



Sits next to **More** in the review screen's bottom bar, so it never covers a card (you can move it to a corner in Stats). It shows your ATP, what one card is worth right now, your flow and how many vesicles are waiting to be opened.

The card under your deck list



Your level, streak and freezes, today's goal, how much of today's due reviews you've done, the daily quests and your last session. Click Mito to open the game. A badge tells you when vesicles are waiting or Fission is ready.

The game window

Open it with the **Powerhouse** button in the top bar, from the deck list card, or **Tools → Powerhouse**. The **?** button in its corner opens the in-game guide. Click Mito to poke him; he has opinions.

Pathways	Upgrades	Vesicles	Enzymes	Wardrobe	Trophies	Stats
Buy metabolism	×2 boosts and study habits	Open your loot	The codex of 48	Mito's looks	56 to earn	Numbers, settings, the guide

Your first five minutes

1. Review a few cards like you always do.
2. Open the game. A short guide walks you through it.
3. Finish the guide and get 3 free vesicles.
4. Open your vesicles. They're full of ATP.
5. Buy your first Glycolysis (15 ATP).

WHILE YOU REVIEW

Fair by design

The rules are built so the game never pushes you to study worse. Grade honestly, go at your own pace, and the ATP follows.

Any button, same ATP

Again pays exactly what Easy pays. The button you press never changes what an answer earns, so grade the way you really did.

Flow, not speed

Each answer in a row adds +0.5% ATP, up to **×1.5** (×1.75 and ×2 with upgrades). More than **3 minutes** between answers resets it (the Pomodoro timer lets it wait out a 5-minute break). Going faster never pays more.

Only real reviews count

Three rules stop mashing and bots. At a normal pace (a couple of seconds a card or more) you'll almost never see them.

Mashing

0.6s ~~0.4s~~ ~~0.5s~~ 4.2s

An answer under a second is fine now and then. It doesn't count if another under-a-second answer came within the 4 answers before it.

One card, all day

1 2 3 4 ~~5~~

A card counts up to **4 times a day**, so looping one card (or a short filtered deck) doesn't pay.

Auto-advance

3.00s 3.01s ×3 ~~3.00s~~

Six answers in a row timed to within 60 ms of each other is a machine, not a person, so from the sixth on they stop counting until the rhythm breaks.

Answers that don't count earn nothing and don't count toward your goal or streak. With pop-ups on, the pill tells you when it happens (at most once every minute and a half for each kind).

Undo is safe

Undoing an answer takes back what it earned: its ATP, its XP and the vesicle it dropped, if you haven't opened it.

Phone reviews count

Reviews from AnkiMobile, AnkiDroid or AnkiWeb are credited once they sync to your computer, even up to 30 days later.

SPEND YOUR ATP

Build Mito's metabolism

Every card starts at 1 ATP, and each pathway you own adds to that for every card from then on. Buy one of a pathway and the next one opens.

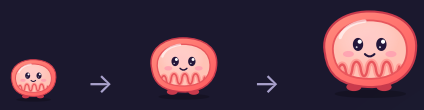
PATHWAY	FIRST ONE COSTS	EACH ADDS*
Glycolysis Glucose → 2 pyruvate for a net 2 ATP. PFK-1 sets the pace.	15	+0.1 / card
Pyruvate dehydrogenase Links glycolysis to the Krebs cycle. Needs B1, lipoic acid, B2, B3 and B5.	270	+0.6
Krebs cycle Each turn makes 3 NADH, 1 $FADH_2$ and 1 GTP.	5,000	+4
Electron transport chain Complexes I, III and IV pump protons. Cyanide and CO block Complex IV.	90K	+25
ATP synthase Complex V spins on the proton gradient. Oligomycin jams the rotor.	1.6M	+160
β-oxidation Long-chain fatty acids enter through the carnitine shuttle.	28M	+1,000
Ketone bodies The brain's main backup fuel in starvation.	500M	+6,300
Creatine phosphate The fastest ATP refill. Runs dry in about ten seconds flat out.	9B	+40K
Slow-twitch fibers Type I muscle: red with myoglobin and packed with mitochondria.	70B	+250K
Mitochondrial biogenesis PGC-1α tells the cell to build more of you.	700B	+1.6M

*Before ×2 boosts and other multipliers. Each one you buy makes the next of that pathway cost 15% more. Use ×10 or Max to buy several at once.

×2 boosts

Own **10, 25, 50 and 100** of a pathway to unlock four upgrades that each **double** it. Each one is a real piece of biochemistry, from fructose-2,6-bisphosphate to the Cori cycle.

Mito grows with them



The more ATP his pathways make, the bigger he gets. He's full size once his pathways alone make about 10 million ATP a card.

Which to buy? Each row in the Pathways tab shows what one more adds per card, what the whole pathway makes now, and its share of your total.

MULTIPLIERS

Upgrades

One-time buys in the Upgrades tab. They unlock as you answer more cards and are bought with ATP. A dot on the tab means you can afford one.

Study habits multiply all your ATP

HABIT	EFFECT	UNLOCKS AT	COSTS
Coffee Adenosine receptor antagonist. You know what you did.	×1.1	150 answers	300
Water bottle Hydrated neurons fire better.	×1.1	500	2,500
Noise-cancelling headphones The library is louder than it looks.	×1.15	1,500	75K
Pomodoro timer 25 on, 5 off: flow now waits out a 5-minute break.	×1.2	3,000	600K
Standing desk Your posture called.	×1.25	6,000	6M
Eight hours of sleep Memories consolidate while you sleep. Seriously, go to bed.	×1.5	10,000	50M
Study group Explaining it out loud is retrieval practice too.	×1.25	20,000	1.5B
Annotated review book Margins full of your own handwriting.	×1.5	35,000	20B
Question bank streak Retrieval practice, now with vignettes.	×1.5	50,000	100B
Spaced repetition mastery You trust the algorithm, and it trusts you.	×2	75,000	600B
Board certified	×2	100,000	1.5T
Attending energy Calm, fast, and still learning.	×2.5	150,000	4T

Vesicle and flow upgrades

Clathrin coat
+15% vesicles
400 answers · 2,000 ATP

SNARE proteins
vesicle ATP ×2
2,000 · 150K

Rab GTPase
rarer vesicles
8,000 · 25M

Dynamin pinch
+15% vesicles
20,000 · 1.5B

Flow state
flow up to ×1.75
1,000 · 20K

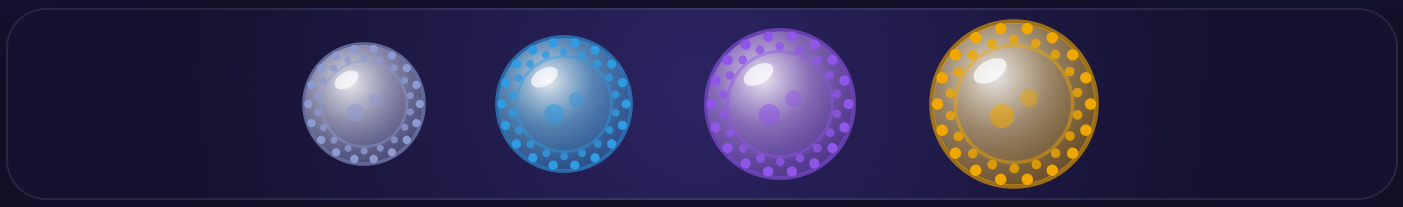
Deep work
flow up to ×2
12,000 · 150M

When Mito divides, all of these stay. Only your ATP, pathways and their ×2 boosts restart.

LOOT

Vesicles

Vesicles bud off while you review, about one every 40 answers that count (more often with the Clathrin coat and Dynamin pinch upgrades). Open them in the game window after your session.



Common	Rare	Epic	Legendary
70% of drops	22% of drops	7.3% of drops	0.7% of drops
ATP 10–20 cards	ATP 30–60 cards	ATP 100–190 cards	ATP 400–750 cards
Enzyme 28%	Enzyme 36%	Enzyme 50%	Enzyme 85%
Outfit 0.6%	Outfit 1.3%	Outfit 2.5%	Outfit 8%
			Freeze 20%

"Cards" means cards' worth of ATP, at what a card was worth when the vesicle dropped (SNARE proteins doubles it). The Rab GTPase upgrade shifts the odds toward rarer vesicles (legendary 1.1%).

Built for 2,000-card days

Drops thin out as a big day goes on, so a heavy day doesn't bury you in vesicles. ATP and XP never thin out.

Full rate	Half	A quarter	An eighth
first 150 answers of the day	answers 151–500	answers 501–1,000	after 1,000

More ways to get them

- 3 to start with, after the first-run guide.
- Each daily quest: a common or rare one. All three: a bonus epic.
- Each trophy: one of the trophy's rarity.
- Each clean sweep and every fifth level: a rare one.

No need to hoard

A vesicle's ATP is set by what a card was worth when it dropped, so waiting doesn't make it bigger. Open them whenever you like; **Open all** does the lot at once.

THE CODEX

Collect the enzymes

48 enzymes in 6 sets, each with a high-yield fact. Every set has 3 common, 2 rare, 2 epic and 1 legendary enzyme. A new set opens with each of your first six titles.

PRE-MED · LEVEL 1
Glucose metabolism
Open from the start

MS1 · LEVEL 5
Glycogen & fat stores
and at least 1 week in

MS2 · LEVEL 10
The mitochondrion
and at least 3 weeks in

STEP 1 SURVIVOR · LEVEL 15
Lipids & steroids
and at least 5 weeks in

MS3 · LEVEL 20
Urea cycle & nucleotides
and at least 8 weeks in

MS4 · LEVEL 25
Heme & neurotransmitters
and at least 12 weeks in

Each set also has that minimum wait after you install, so on a busy schedule it can arrive a little after its title.

Every legendary is a rate-limiting enzyme

- **PFK-1** — Rate-limiting step of glycolysis. Fired up by AMP and fructose-2,6-bisphosphate.
- **Glycogen phosphorylase** — Rate-limiting for glycogenolysis. Needs vitamin B6.
- **Isocitrate dehydrogenase** — Rate-limiting for the Krebs cycle. Activated by ADP; inhibited by ATP and NADH.
- **HMG-CoA reductase** — Rate-limiting for cholesterol synthesis. Statins block it.
- **Carbamoyl phosphate synthetase I** — Rate-limiting for the urea cycle. Needs N-acetylglutamate to switch on.
- **ALA synthase** — Rate-limiting for heme synthesis. Needs vitamin B6.

Duplicates raise expression

More enzyme, higher Vmax, same Km. Each level above 1 adds +0.25% ATP, and every duplicate also pays some ATP. Copies needed for each level:

	LV 1	2	3	4	5
Common	1	3	6	12	20
Rare	1	3	5	9	15
Epic	1	2	4	7	10
Legendary	1	2	3	4	6

Set bonuses

Each complete set: **+5% ATP**. All six: **+10%** more on top.

Bad luck has a limit

You never get more than **2 duplicate legendaries in a row** while you're still missing one. The codex tells you when the next one is guaranteed new.

EVERY DAY

Goal, streak, quests

Your day follows Anki's own (it starts at Anki's "next day starts at" hour, 4 a.m. unless you changed it).

Daily goal and streak

Answer your **daily goal** (100 cards to start; change it in Stats) to keep your streak going. Reaching it also pays a **knowledge dividend**: 1% of your mature cards, in cards' worth of ATP (up to 1,000).

A clean sweep counts too: finish every due review and the day counts for your streak even below your goal, as long as at least half of the day's answers counted.

Streak freezes

Every **7 streak days** earns a freeze. You can bank 2, and a lucky legendary vesicle can add one more. A missed day uses one automatically, so one bad day doesn't end a long streak.



Three daily quests

Sized to how much you usually study, new at rollover.

- [x] Answer 60 cards — common
- [ ] Answer 125 cards — rare
- [ ] Reach a flow of 50 — rare

The third one changes: a flow target, cards before noon, two separate sessions, finding 2 vesicles, or a clean sweep. Finish all three for a bonus epic vesicle.

Trophies

56 of them, each worth **+1% ATP** forever and a vesicle. A few to aim for:

- **Boards eve**: 2,000 cards in one day
- **Centurion**: a 100-day streak (unlocks the halo)
- **Early bird**: 50 cards before 8 a.m.
- **Inbox zero**: clean sweeps on 7 days
- **Mitochondrial Eve**: divide five times
- ...and one secret one.

XP, levels and titles

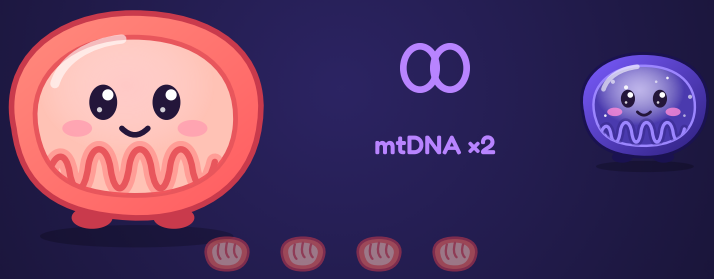
Every answer that counts is 1 XP. Each level adds **+1% ATP**, every fifth level gives a rare vesicle, and some levels unlock looks for Mito (a scrub cap at 3, round glasses at 5, a hospital badge at 10, a stethoscope at 15, the Gram stain slide at 25 and a graduation cap at 30).

1 Pre-med · **5** MS1 · **10** MS2 · **15** Step 1 survivor · **20** MS3 · **25** MS4 · **30** Intern · **35** Resident · **45** Chief resident · **55** Fellow · **65** Attending · **80** Department chair · **100** Nobel laureate

THE LONG GAME

Mito grows, then divides

Mito grows as his pathways make more ATP. Once you've made 10 billion ATP in total, he can divide, like real mitochondria. It's optional, like Cookie Clicker's Ascension.



✕ Restarts

Your ATP, Mito's pathways and their ×2 boosts. He starts small again.

✓ Keeps

Study habits, vesicle and flow upgrades, level, streak, trophies, enzymes, looks, unopened vesicles and every mtDNA copy.

∞ mtDNA: +25% ATP per copy, forever

Your copies grow with the cube root of all the ATP you've ever made, so each one takes more than the last:

1	2	3	10	100
at 10 billion	at 80 billion	at 270 billion	at 10 trillion	at 10 quadrillion

mtDNA is real: it encodes 13 proteins of the electron transport chain and ATP synthase. More copies, more ATP.

✦ Why divide?

Each copy of a pathway costs 15% more than the one before, so growth slows down. Dividing trades the pathways for a permanent bonus, and you rebuy them far faster, so each climb goes further than the one before.

✓ When?

A **Fission is ready** card appears under Mito once dividing would gain at least 3 copies and at least double what you have. You can also divide earlier from Stats.

Divisions unlock looks: the Galaxy skin at your first and the Holographic skin at your fifth. Past generations float in the background, and Mito's name shows his generation.

QUESTIONS

Good to know

The things people ask most.

Will it mess with my cards or AnkiHub?

No. Powerhouse only reads your review history. It never changes cards, scheduling, decks or sync, so it can't affect AnkiHub, AnKing decks or other add-ons.

Does it work on my phone?

Powerhouse runs in Anki on a computer (version 23.12 or newer). Reviews on AnkiMobile, AnkiDroid or AnkiWeb still count once they sync to that computer.

I do 1,000+ cards a day. Is that OK?

Yes. Vesicle drops thin out over a big day, each enzyme set has a minimum wait after install so the codex lasts, and there's a trophy for 2,000 in a day.

Why did an answer earn nothing?

It was one of the fair-play cases: a burst of sub-second answers, a card's 5th answer of the day, or a machine-steady rhythm. Normal reviewing is almost never affected.

I use two computers.

The game lives on each computer, so each one keeps its own game. Reviews you do anywhere still count on each once they sync.

Where's my progress saved?

In the add-on's **user_files** folder, with a daily backup. Anki keeps it when the add-on updates, and it follows your profile if you rename it.

Can I turn things off?

Everything is in Stats: daily goal, where the pill sits (or hide it), pop-ups, the deck list card, the top-bar button, sounds and reduced motion.

Is it ever finished?

Not quickly. Roughly how long things take, from simulations of the real game rules:

	250 CARDS A DAY	1,500 CARDS A DAY
Every pathway	**about 6 months**	**under 2 months**
The whole codex (48 enzymes)	**about 6 months**	**about 3 months**
First Fission worth it	**about 3½ months**	**about 1 month**
Then a division about every	**6 weeks**	**2–3 weeks**
The rarest outfits	**a year or more**	**several months**

✦ **Found a bug or have an idea?** Use the support link on the add-on's AnkiWeb page. Include your Anki version (Help → About) and, if you can, **user_files/powerhouse.log**.